Applicability analysis of large language models for AI-based concrete mix design

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Abstract

In the field of concrete compressive strength prediction and mix design, various studies have been conducted focusing on machine learning (ANN, SVR, etc.), but research applying Large Language Models (LLMs) is relatively limited. Therefore, in this study, the effects of learning characteristics between AI models on mix design results were analyzed through compressive strength and CO_2 emissions by comparing LLMs with existing previous research approaches (ANN, SVR). As a result of analyzing the 7-day compressive strength and CO_2 emissions for the mix proportions derived based on AI models, the compressive strengths of the ANN, SVR, and LLM-based mix proportions were evaluated as 5.38 MPa, 13.66 MPa, and 15.46 MPa, respectively, and the CO_2 emissions were calculated as 320, 260, and 210 $kgCO_2e/m^3$, respectively. Consequently, the LLM-based mix design showed a low strength error rate (3.1%) and the lowest CO_2 emissions (210 $kgCO_2e/m^3$), confirming that it can derive mix proportions that satisfy design goals more accurately than existing ANN and SVR models.

1 Introduction

Concrete is a representative structural material that can simultaneously ensure structural safety, durability, and economic efficiency, and is widely utilized throughout civil and architectural structures [1, 2]. The quality of concrete is influenced by the types of mixing materials, mixing proportions, and construction and curing conditions [3, 4]. While predicting concrete compressive strength has traditionally been performed through experimental verification, research has been continuously conducted to quantitatively analyze the relationship between mix variables and compressive strength using regression analysis techniques alongside the accumulation of experimental data [5, 6]. Since then, as machine learning techniques were introduced, research has expanded to learn complex patterns from large-scale data and effectively model the nonlinear relationship between mix variables and compressive strength [7-9].

Specifically, from the late 1990s to the early 2000s, numerous studies on compressive strength prediction utilizing various material properties and mixing proportions as input variables were reported, centered on Artificial Neural Networks (ANN). Subsequently, Support Vector Regression (SVR) has also been widely utilized in terms of prediction accuracy and generalization performance [10-13]. In contrast, research applying Large Language Models (LLMs) to concrete compressive strength prediction and mix design has not been sufficiently and systematically conducted compared to ANN and SVR-based approaches. Existing studies have mainly focused on predicting compressive strength based on given mix conditions, and research on inverse design approaches that directly derive mix proportions satisfying target compressive strengths, or multi-objective optimization considering both strength performance and environmental factors, has been reported relatively limitedly [14, 15].

Accordingly, in this study, concrete compressive strength prediction and mix design are performed by applying ANN, SVR, and LLM. While ANN and SVR are prediction-oriented machine learning models that learn the relationship between numerical input variables and outputs, LLMs differ in their approach as models capable of comprehensively interpreting and reasoning through various conditions and constraints based on a language-based learning structure. Unlike the existing machine learning-based prediction model-centered approaches, the significance of this study lies in extending concrete mix design to an inverse design problem by applying a conversation-based reasoning model to the process of deriving mix proportions that satisfy target compressive strengths, and systematically comparing and analyzing the application characteristics and distinctions between AI models.

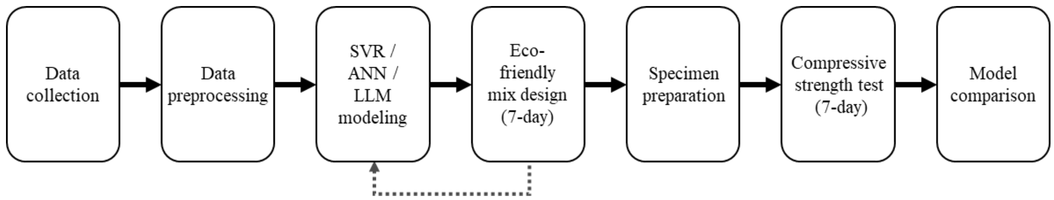

Figure 1 Flow chart

2 AI modeling framework

2.1 LLM

LLMs are language-based AI models capable of analyzing and utilizing meaning and context by processing large-scale unstructured language data [16]. LLMs are provided as AI services by various companies; representative models include Anthropic's Claude, OpenAI's ChatGPT, Google's Gemini, Microsoft's Copilot, and Meta's Llama. In this study, the model characteristics and context windows of representative LLMs—Claude, ChatGPT, and Gemini—were compared to select the model best suited for the research objectives, as summarized in Table 1.

Table1. Comparative analysis of AI models

Category	Claude Sonnet 4.5	GPT-5.2	Gemini 3
Developer	Anthropic	OpenAI	Google
Release date	September 2024	December 2025	December 2025
Key features	• Structuring existing data • Detailed explanation of concepts	• Reorganization of existing data • Strong storytelling capability	• Factual and contextual understanding • Extraction of key data characteristics • Feature-based creative idea generation
Limitations	• Lack of response consistency• Overly verbose explanations	• Limited precision in some tasks • Strong bias toward storytelling	• Relatively weak storytelling capability
Context window	• Below 4 K tokens	• 100 K tokens (high precision) • Operable beyond 200 K with performance degradation	• Stable at 250 K tokens • Usable up to 500 K tokens • Responses possible up to 800K with reduced accuracy

The context window refers to the range of text a model can simultaneously consider during a single inference process, playing a crucial role in integrating and analyzing large-scale research data or long-form information within a single context [17]. The context window varied by model, and performance degradation has been reported in some models as input length increases [18]. Therefore, to effectively analyze large-scale experimental data, this study selected Gemini as the LLM, as it demonstrates relatively stable performance in an expanded context window environment. Furthermore, based on this LLM, existing mix data and carbon emission information will be analyzed within a single context to examine the feasibility of deriving new eco-friendly concrete mix proportions.

2.2 ANN

An ANN is an information processing system that mimics the biological nervous system, consisting of numerous neurons connected in a complex network [19]. ANNs can identify complex correlations and nonlinearities between input and output data [20]. The Multi-Layer Perceptron (MLP) is a representative neural network structure for modeling such nonlinear relationships [20]. This model belongs to the Feedforward Neural Network (FNN) category, where information is passed from the input layer to the output layer in a forward direction [21]. Structurally, it consists of an input layer, one or more hidden layers, and an output layer, with each layer composed of multiple neurons [22]. These neurons form a network of weighted connections. For neural network training, the Levenberg–Marquardt (LM) algorithm—a second-order algorithm with fast convergence characteristics in the weight optimization process to minimize the error function—is utilized [23].

Such multi-layer structures and weight-based learning mechanisms provide the structural foundation for ANNs to approximate complex nonlinear functions. This capability is suitable for modeling the relationship between concrete mix proportions and compressive strength, where multiple variables such as cement, water, aggregates, and SCMs interact. However, in concrete mix design, "inverse design"—deriving material combinations that satisfy a target compressive strength—is a critical challenge. As a forward model that predicts output from input, an ANN has limitations in direct application to inverse design because an inverse function is not defined (multiple input combinations can exist for the same output). To overcome this, this study defined the difference between the target strength and the ANN-predicted value as the objective function. To minimize this objective function, a constrained optimization technique was applied to iteratively update and minimize the function for continuous mix variables. This implemented an inverse design procedure that searches for mix proportions approaching the target strength while maintaining the ANN's forward prediction characteristics.

2.3 SVR

In this study, Support Vector Regression (SVR) was utilized, which extends the principles of Support Vector Machines (SVM) to regression analysis. SVR maintains excellent predictive performance even with small datasets and exhibits robustness against noise [24]. To ensure the generalization performance of the regression model, SVR introduces an ϵ-insensitive loss function to the SVM-based regression framework [25]. In this approach, if the difference between the predicted value derived from the linear estimation function $f(x)=(w\cdot\emptyset(x))+b$ and the actual value is within the precision parameter ϵ, it is not recognized as an error. Through this process, the weight vector (w) and the constant (b) are optimized to minimize the regularized risk function, thereby enhancing the model's robustness [26].

Furthermore, a Kernel function was applied to effectively model complex nonlinear relationships between input variables, allowing for the precise prediction of continuous numerical data such as compressive strength and the mapping of data into a high-dimensional space. Consequently, SVR maintains stable predictive performance even with limited experimental datasets and possesses high tolerance for data noise that may occur during the experimental process. While ANNs have strengths in learning complex patterns based on large-scale data, SVR prevents overfitting and provides high predictive reliability even when the number of data points is limited [27, 28]. Additionally, to minimize the objective function for inverse design, a constrained optimization technique was applied, similar to the approach used for the ANN.

3 Experiments and results

3.1 Experiment overview

The 7-day compressive strength was set to 15 MPa by inversely estimating the strength development relationship based on a reference 28-day compressive strength of 23 MPa, which represents commonly used ordinary concrete in construction practice [29]. Based on this target, this study applied LLM, ANN, and SVR models to generate concrete mix proportions satisfying a target 7-day compressive strength of 15 MPa and compared the results across models. All models utilized identical mix proportion–

compressive strength datasets, and the generated mix proportions were validated through actual concrete specimen production and compressive strength testing. The experimental procedure consisted of three stages: data collection and preprocessing, model training and mix generation, and specimen preparation and strength testing. ANN and SVR models were trained to predict 7-day compressive strength using the collected data, while the LLM generated mix proportions based on the same data information. When multiple mix proportions satisfying the same target compressive strength were generated, the CO_2 emissions for each mix were calculated using material-specific CO_2 emission factors, and the mix with the relatively lowest CO_2 emissions was selected as the final experimental candidate [30–36].

3.2 Data collection and preprocessing

In this study, concrete mix data were collected based on mix proportion–compressive strength datasets reported in previous experimental studies. A total of 2,322 mix–strength data points were compiled, with the primary sources being experimental research articles published on ScienceDirect and datasets released on Mendeley Data [37–57]. All data were based on experimentally measured 7-day compressive strength values. The collected data consisted of mix proportions of cement, water, fine aggregate, coarse aggregate, supplementary cementitious materials (SCM), and admixtures, along with the corresponding 7-day compressive strength. Data that fell outside the mix design range were excluded, and the remaining data were integrated into a single unified database. The basic statistical statistics of the integrated database are presented in Table 2.

Table 2. Descriptive statistics of detailed concrete mixture variables and compressive strength

Variables		Mean	Max	Min	Std.
Cement (kg/m^3)		341.18	650.00	112.00	94.58
Water (kg/m^3)		174.60	314.30	116.00	20.47
FineAgg (kg/m^3)		728.64	992.60	400.00	94.15
CoarseAgg (kg/m^3)	Crushed aggregate	1005.54	1250.00	266.40	149.83
	Crushed granite aggregate	979.39	987.00	974.00	3.67
	Foamed waste glass aggregate	471.00	471.00	471.00	0
	Lightweight aggregate	748.70	748.70	748.70	1.25e-13
	Natural agregate	1030.28	1411	0	193.784
	Recycled coarse aggregate	25	25	25	0
SCM (kg/m^3)	Flyash	86.32	329.00	0.27	41.99
	BFS	116.67	359.40	0.03	86.15
	Metakaolin	48.80	58.80	38.80	10.29
	Silica fume	42.35	141.00	7.85	29.30
Admixture (kg/m^3)	Superplasticizer	9.53	64.80	0.01	8.95
	High-range air-entraining water-reducing agent	5.18	25.13	0.07	4.908

7days compressive strength (MPa)	36.53	87.00	3.20	14.65

Note: FineAgg = Fine Aggregate CoarseAgg = Coarse Aggregate, SCM = Supplementary Cementitious Materials.

The input variables were divided into continuous variables and categorical variables. The continuous variables are defined as variables representing the usage amounts of each material expressed in kg/m^3, including cement, water, fine aggregate, coarse aggregate, additional aggregate, SCM, additional SCM, and admixtures. The categorical variables consist of code variables indicating material types or usage presence. The output variable was defined as the 7-day compressive strength (MPa). Derived variables with physical meaning were added so that the model could effectively learn the relationship between mix proportions and compressive strength. The binder content was defined as the sum of cement and all SCMs, based on which the water-to-cement ratio (W/C) and water-to-binder ratio (W/B) were calculated. In addition, the SCM replacement ratio (SCM_ratio) was included as an input variable. The basic statistics of the derived variables are presented in Table 3. For both ANN and SVR models, the entire dataset was divided into training, validation, and testing sets at ratios of 70%, 15%, and 15%, respectively.

Table 3. Descriptive statistics of calculated mix proportion parameters

Derived variables		Mean	Max	Min	Std.
Binder (kg/m^3)	Cement (kg/m^3) + SCM (kg/m^3) + Add_SCM (kg/m^3)	399.00	716.96	183.00	92.41
W/C	Water (kg/m^3) / Cement (kg/m^3)	0.55	1.15	0.20	0.17
W/B	Water (kg/m^3) / Binder (kg/m^3)	0.46	1.00	0.18	0.12
SCM_ratio	(SCM (kg/m^3) + Add_SCM (kg/m^3)) / Binder(kg/m^3)	0.14	0.70	0.00	0.14

3.3 Model-based mix proportion derivation method

In the mix proportion generation stage, continuous mix variables were defined as decision variables adjusted to satisfy the target compressive strength. The decision variables consisted of the material usage amounts (kg/m^3) of cement, water, fine aggregate, coarse aggregate, SCM, and admixtures, whereas categorical code variables indicating material types or usage presence were treated as fixed condition variables during the mix proportion generation process. Identical physical constraints were applied to all models to ensure that the generated mix proportions remained within the practical concrete mix design range. The allowable constraint ranges for mix proportion optimization are presented in Table 4. The continuous mix variables were set to be explored within the minimum–maximum ranges observed in the training data.

Table 4. Range of allowable constraints for mixture optimization

Category	Constraint	Min	Max
W/C	Water_kgm3 / Cement_kgm3	0.35	0.80
W/B	Water_kgm3 / Binder	0.30	0.70
Total coarse aggregate content (kg/m^3)	CoarseAgg + Add_CoarseAgg	750	1300
Total SCM content (kg/m^3)	SCM + Add_SCM	0	250
Maximum aggregate size (mm)	CoarseAgg_Max_Size	10	25

LLM-based mix proportion generation was performed by providing the preprocessed mix proportion–compressive strength data and physical constraint conditions in textual and tabular forms, through which candidate mix proportions satisfying the target 7-day compressive strength were generated.ANN- and SVR-based mix proportion generation was carried out using an inverse design approach based on the trained compressive strength prediction models. In this process, constrained optimization techniques were applied to adjust the continuous mix variables such that the error between the predicted 7-day compressive strength and the target value was minimized. When multiple mix proportions satisfying the same target compressive strength were generated, CO_2 emissions for each mix were calculated using CO_2 emission factors, and the mix with the relatively lowest CO_2 emissions was selected as the final experimental candidate.

Table 5 Emission factors of concrete materials

Type	EF	Type	EF	Type	EF
OPC	0.85	Water	0.0003	Crushed agg.	0.005
Crushed granite agg.	0.005	Natural agg.	0.005	RCA	0.002
LWA	0.30	FWGA	0.15	SP	2.0

Note: OPC = Ordinary Portland Cement, agg = Aggregate, RCA = Recycled Concrete Aggregate, LWA = Lightweight Aggregate, FWGA = Fine Waste Glass Aggregate, SP = Superplasticizer, EF = Emission Factor ($kgCO_2e/kg$)

3.4 Model training results

During the training process of the ANN model, the validation error was evaluated based on the Mean Squared Error (MSE) of the validation dataset, and the network at the point where the validation error was minimized was selected as the optimal model. The training was conducted for up to 50 epochs, with an early stopping mechanism applied based on the validation error. As a result, the validation performance was found to be most superior at 30 epochs, after which the training was terminated to prevent overfitting.

According to the regression analysis, the correlation coefficients (R) for the training, validation, and testing datasets were 0.94, 0.92, and 0.93, respectively. The overall R-value for the entire dataset was confirmed to be 0.93, demonstrating high predictive performance. Furthermore, the evaluation of the predictive performance on the test dataset revealed that the Root Mean Square Error (RMSE) of the ANN model was 5.79.

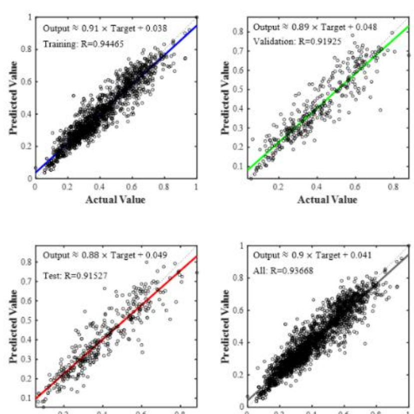

Figure 2 (A) Regression plots of the ANN

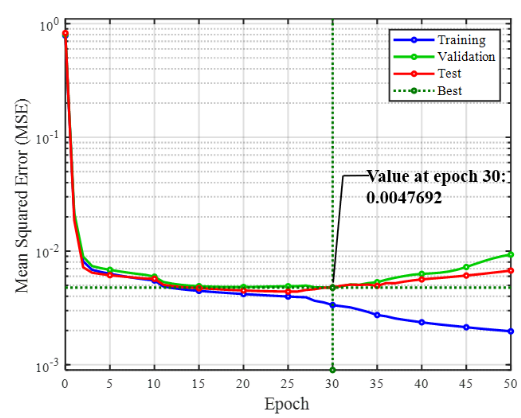

Figure 2 (B) MSE versus epoch for the ANN

To enhance the predictive accuracy of the SVR model, hyperparameter tuning was performed based on Bayesian optimization, and the optimal combination of hyperparameters was derived through a total of 60 evaluations. As a result, the correlation coefficients (R) for the training and testing data were found to be 0.89 and 0.85, respectively. Furthermore, the predicted values exhibited a distribution concentrated around the identity line ($y = x$), indicating a high degree of consistency with the measured values. In particular, superior predictive performance was observed within the range of 10–60 MPa.

These results suggest that the SVR model stably learned the strength development mechanism across both the training and testing datasets. This implies that the model has secured sufficient reliability for practical application in concrete mix design, based on its high predictive precision and generalization performance.

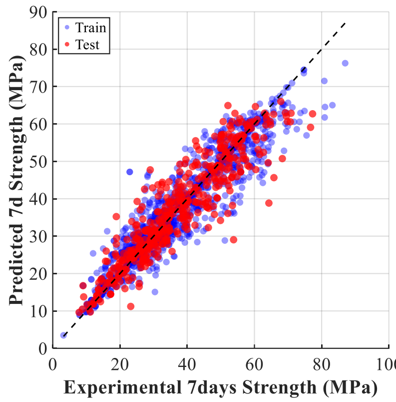

Figure 3 (A) SVR regression plot (Train–Test).

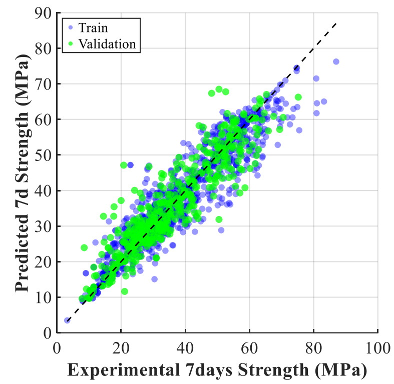

Figure 3 (B) SVR regression plot (Train–Validation).

In the case of the LLM, the model was utilized after a training process that involved providing step-by-step conditions and incorporating feedback to systematically examine the relationships between concrete mix design, strength, and environmental impact. While iteratively adjusting the eco-friendly concrete mix design, the water-to-binder (W/B) ratio, binder content, and SCM replacement rate were established as the primary variables. Meanwhile, air content and slump were assumed to be at laboratory standard conditions to ensure experimental feasibility and reproducibility. The generated candidate mixes were verified to ensure that the key mix variables remained within the range of the existing dataset. By comprehensively considering these criteria, a single optimal mix ratio was derived consisting of 130 kg/m³ of cement and 130 kg/m³ of fly ash, which satisfied the constraints while exhibiting the lowest uncertainty. Furthermore, the actual material quantities for the experimental specimens were calculated and presented in accordance with the specimen dimensions (Ø 100×200 mm).

3.5 Specimen preparation and compressive strength testing

In this study, concrete specimens were fabricated using the final mix ratios derived from each model. The specimens were cured for 7 days under identical conditions, after which compressive strength tests were conducted in accordance with KS F 2405. The material composition and quantities for each model-based mix ratio are presented in Table 6. By comparing the model predictions with the experimental results of the 7-day compressive strength, the practical applicability of the mix ratios generated by each model was evaluated. Furthermore, the calculation of CO_2 emissions for the AI-based design mix ratios revealed that the total CO_2 emissions for the ANN, SVR, and LLM-based mixes were approximately 320, 260, and 210 $kgCO_2e/m^3$, respectively. Notably, the LLM-based mix ratio exhibited the lowest CO_2 emission characteristics, demonstrating superior environmental performance.

Table 6. Material composition and quantities of model-specific mix designs

Model	C (kg/m3)	W (kg/m3)	FA (kg/m3)	CA_type	CA (kg/m3)	SCM _type	SCM (kg/m3)	ADM _type	ADM (kg/m3)
LLM	216.45	180.38	901.88	Crushed Aggregate	1082.25	BFS	93.37	AE	1.55
ANN	146.42	116.71	409.56	Crushed Aggregate	1071.64	BFS	29.71	None	0
SVR	214.33	148.54	993.13	Crushed Aggregate	1080.13	None	0	None	0

3.6 Compressive strength results by model

Table 7 presents the maximum 7-day compressive strength for the mix ratios derived by each model, and Figure 4 illustrates the representative stress–strain curve results for each model.

Table 7. 7-day maximum compressive strength by model

7-day maximum compressive strength (MPa)			
	ANN	SVR	LLM
Specimen 1	3.91	14.19	15.98
Specimen 2	6.26	12.91	15.01
Specimen 3	5.96	13.87	15.40

Based on the maximum 7-day compressive strength results presented in Table 7, the average compressive strengths calculated for each model were confirmed to be 5.38 MPa for ANN, 13.66 MPa for SVR, and 15.46 MPa for LLM. Furthermore, the error rates relative to the target compressive strength were found to be 64.1%, 8.9%, and 3.1%, respectively.

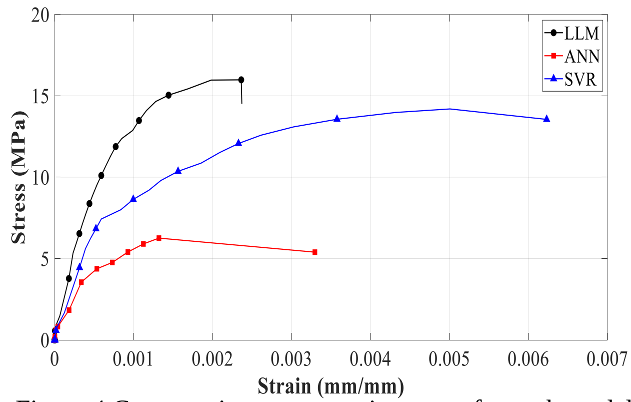

Figure 4 Compressive stress–strain curves for each model

Consequently, LLM demonstrated the most superior performance in terms of reproducing the target compressive strength, while the SVR-based mix design exhibited predictive performance relatively close to the target value. In contrast, the ANN-based mix design showed relatively limited performance in deriving a mix ratio that meets the target compressive strength level.

4 Discussion

The comparative analysis of ANN, SVR, and LLM models revealed that the LLM model reached a maximum stress of 15.98 MPa at a strain of 0.00236, indicating its ability to derive a mix design with relatively high stiffness and load-resistance characteristics. This suggests that the LLM can more precisely learn the nonlinear and complex correlations between mix design factors compared to conventional regression-based models, thereby effectively reflecting the mechanical properties of concrete materials.

In conclusion, this study demonstrates that the LLM shows a certain level of improvement in compressive strength prediction performance over existing machine learning models. Based on these findings, this research holds significance as a foundational study for the applicability of data-driven concrete mix design and its potential for practical field utilization.

5 Conclusions

In this study, a total of 2,322 concrete mix data points were established as a training dataset to train LLM, SVR, and ANN models, and the optimal mix variables were quantitatively derived. Furthermore, the eco-friendliness of the concrete mixes was quantitatively evaluated based on the summation of eco-friendly coefficients during the mix design process, and the results are as follows.

- The training results of the compressive strength prediction models based on ANN, SVR, and LLM showed predictive performances of 0.93 (ANN) and 0.85 (SVR). Meanwhile, the LLM performed target-performance-oriented mix ratio generation by reflecting the physical relationships between mix variables, even without separate numerical-based regression training.
- The results of the 7-day compressive strength tests for the design mix ratios derived from the AI models showed strengths of 5.38 MPa (ANN), 13.66 MPa (SVR), and 15.46 MPa (LLM). Among these, the LLM demonstrated the result closest to the target compressive strength of 15 MPa.
- The calculation of CO_2 emissions for the AI-based design mix ratios showed total emissions of approximately 320, 260, and 210 $kgCO_2e/m^3$ for the ANN, SVR, and LLM-based mixes, respectively. Among them, the LLM-based mix ratio showed the most superior performance in terms of environmental sustainability.

This study empirically compared and analyzed the performance of existing SVR and ANN models against the LLM for data-driven, objective concrete mix design, thereby verifying the novel applicability of the LLM. In particular, by performing eco-friendly optimization that incorporates carbon emission factors into the design process, this research provides a technical basis for multi-objective mix design that simultaneously achieves strength performance and carbon reduction, yielding results that align with ESG management requirements.

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